

$$\equiv \forall x \forall y ((x=y) \vee \neg C(x,y) \vee \exists z (((x \neq z) \wedge C(x,z)) \wedge (z \neq y)))$$

(Double-negation Law)

For any arbitrary pair of students, either both are the same student or they have not chatted with each other, or there exists atleast one other student in the class who has chatted with one of them.

c) No student in this class has sent email to exactly two other students in this class.

$E(x,y)$: x has sent email to y . Common domain: Students of the class

$$\forall x \exists y \exists z \neg ((x \neq y) \wedge (x \neq z) \wedge (y \neq z) \wedge E(x,y) \wedge E(x,z) \wedge \forall w ((x \neq w) \wedge E(x,w) \Rightarrow ((w=y) \vee (w=z))))$$

$$\begin{aligned} & \neg \forall x \exists y \exists z \neg ((x \neq y) \wedge (x \neq z) \wedge (y \neq z) \wedge E(x,y) \wedge E(x,z) \wedge \forall w ((x \neq w) \wedge E(x,w) \Rightarrow ((w=y) \vee (w=z)))) \\ & \Rightarrow ((w=y) \vee (w=z))) \equiv \exists x \neg \exists y \exists z \neg ((x \neq y) \wedge (x \neq z) \wedge (y \neq z) \wedge E(x,y) \wedge E(x,z) \wedge \forall w ((x \neq w) \wedge E(x,w) \Rightarrow ((w=y) \vee (w=z)))) \quad (\text{De Morgan's law of quantifiers}) \\ & \equiv \exists x \forall y \neg \exists z \neg ((x \neq y) \wedge (x \neq z) \wedge (y \neq z) \wedge E(x,y) \wedge E(x,z) \wedge \forall w ((x \neq w) \wedge E(x,w) \Rightarrow ((w=y) \vee (w=z)))) \quad (\text{De Morgan's law of quantifiers}) \\ & \Rightarrow ((w=y) \vee (w=z))) \quad (\text{De Morgan's law of quantifiers}) \\ & \equiv \exists x \forall y \forall z \neg \neg ((x \neq y) \wedge (x \neq z) \wedge (y \neq z) \wedge E(x,y) \wedge E(x,z) \wedge \forall w ((x \neq w) \wedge E(x,w) \Rightarrow ((w=y) \vee (w=z)))) \quad (\text{De Morgan's law of quantifiers}) \\ & \Rightarrow ((w=y) \vee (w=z))) \quad (\text{De Morgan's law of quantifiers}) \\ & \equiv \exists x \forall y \forall z ((x \neq y) \wedge (x \neq z) \wedge (y \neq z) \wedge E(x,y) \wedge E(x,z) \wedge \forall w ((x \neq w) \wedge E(x,w) \Rightarrow ((w=y) \vee (w=z)))) \quad (\text{Double negation law}) \\ & \Rightarrow ((w=y) \vee (w=z))) \end{aligned}$$

There exists atleast one student in the class who has sent emails ^{to} exactly two other students in this class.